

**Forecasting Ontario Residential Building-Permit Values Using Explainable Time-Series
and Machine-Learning Models**

Synopsis

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Introduction

Background and Context

Construction activity has broad implications for economic growth, employment, municipal infrastructure, housing supply, financing, and the demand for building materials. Decisions made by builders, suppliers, lenders, workforce planners, and municipal economic-development teams depend on an accurate understanding of near-term construction intentions. Building permits are particularly useful for this purpose because they are issued before most on-site construction begins. Statistics Canada describes its monthly Building Permits Survey as a leading indicator of residential and non-residential construction activity and uses the resulting estimates as inputs to national investment and capital-stock measures (Statistics Canada, 2026b).

The proposed study focuses on residential building-permit values in five Ontario census metropolitan areas (CMAs): Toronto, Hamilton, Kitchener-Cambridge-Waterloo, London, and Windsor. Ontario is an appropriate setting because its regions differ in population scale, housing-market structure, industrial composition, and sensitivity to credit conditions. A pooled regional design provides substantially more analytic observations than a single provincial series while preserving an operationally meaningful level of geography. The primary source, Statistics Canada Table 34-10-0292-01, contains monthly permit values beginning in 2018 and supports filters for geography, seasonal adjustment, value type, structure, and type of work (Statistics Canada, 2026a).

Prior research supports the use of economic indicators and time-series methods for construction-demand forecasting. Goh (1996) found that economic indicators could be used to forecast residential construction demand and that a nonlinear model improved accuracy over multiple regression in the study setting. Goh (2000) later showed that hybrid optimization

could further reduce error. Sing et al. (2015) demonstrated the relevance of macroeconomic indicators in forecasting private-sector construction work, while Cheung and Granovsky (2016) confirmed that building permits have predictive content as a leading housing and economic indicator in Canada. Coble and Pincheira (2017) further showed that building-permit forecasts can improve when timely exogenous information is added to conventional benchmarks.

Despite this evidence, a technically sophisticated model is not automatically useful. Construction series are seasonal, serially correlated, subject to extreme project values, and vulnerable to structural breaks. Forecasting research therefore emphasizes credible out-of-sample validation, comparison with simple benchmarks, and metrics that are interpretable across differently scaled series (Bergmeir et al., 2018; Hyndman & Koehler, 2006; Makridakis et al., 2018; Tashman, 2000). The proposed capstone responds to these requirements by comparing transparent statistical baselines with tree-based machine-learning models, using rolling-origin validation and explainability methods to convert predictions into decision-relevant evidence.

Problem Statement

Ontario construction stakeholders can access historical permit information, but the public data are primarily descriptive and do not directly provide reproducible, region-specific, next-month forecasts or explanations of the variables driving those forecasts. Planning based only on the latest observation or informal judgment may be distorted by seasonality, temporary shocks, and the unusually large value of individual projects. The project addresses this gap by developing and validating an explainable forecasting framework that estimates next-month residential building-permit values for selected Ontario CMAs, tests whether publicly available macroeconomic indicators add predictive value, compares statistical and machine-learning approaches against strong baselines, and identifies where model performance is sufficiently reliable to support planning.

Purpose of the Study

The purpose of this quantitative secondary-data study is to forecast monthly residential building-permit values, explain the temporal and macroeconomic drivers associated with those forecasts, compare alternative model families, and translate the results into an operational decision-support framework. The study is predictive rather than causal. It will not claim that a change in interest rates or unemployment directly causes a specific change in permit values. Instead, it will test whether lagged permit history, calendar variables, the Bank of Canada target for the overnight rate, and Ontario unemployment improve out-of-sample prediction. The final product will include a reproducible data pipeline, model-comparison results, regional error profiles, explainability outputs, and recommendations for responsible use.

Scope and Objectives

Study Scope

The unit of analysis will be a CMA-month. The planned period is January 2018 through May 2026, which yields 101 months for each of five CMAs before feature engineering. The primary outcome is the next month's seasonally adjusted residential building-permit value in thousands of current Canadian dollars. The main analysis will retain the total residential structure and total type-of-work categories to prevent an unnecessarily large number of subgroup models. A sensitivity analysis will repeat the preferred model using constant-dollar values when the source table provides the required series. The principal forecast horizon is one month because it is operationally useful and can be evaluated with many rolling forecast origins within a 10-week capstone.

The predictor set is deliberately limited. It will include permit lags at one, three, six, and 12 months; rolling means and volatility measures; month-of-year encodings; CMA indicators; the Bank of Canada target for the overnight rate; the Ontario unemployment rate; and a prespecified pandemic-period indicator. No personally identifiable information, addresses, permit applicants, or protected administrative records will be used. All data are

public, aggregated government statistics. The project therefore presents minimal privacy risk, although the learner will follow Walsh College requirements for any formal determination regarding institutional review.

Objectives

Describe and quantify the trend, seasonality, persistence, volatility, and structural changes in residential building-permit values across the five selected Ontario CMAs.

Determine whether macroeconomic variables add incremental predictive value beyond permit history and calendar effects.

Compare seasonal-naïve, regularized regression, random forest, and extreme gradient boosting forecasts under a leakage-resistant rolling-origin validation design.

Explain the preferred model, evaluate its reliability by CMA, and develop recommendations for responsible operational use.

Research Question 1 (RQ1)

RQ1: What trend, seasonal, persistence, and volatility patterns characterize monthly residential building-permit values across the selected Ontario CMAs? The associated hypothesis is that lagged permit values and annual seasonal terms have statistically significant relationships with the next-month value. All constructs are measurable through government-reported monthly values and derived lag or calendar variables.

Research Question 2 (RQ2)

RQ2: Do the Bank of Canada target for the overnight rate and the Ontario unemployment rate provide statistically and practically significant incremental predictive information beyond permit history, seasonality, and CMA effects? The hypothesis is that adding the two macroeconomic predictors will reduce rolling-origin mean absolute error by at least 5% relative to an otherwise identical autoregressive model. The 5% threshold is prespecified as a practical, not merely statistical, improvement criterion.

Research Question 3 (RQ3)

RQ3: Which candidate model produces the most accurate and stable one-month-ahead forecasts across time and regions? The hypothesis is that at least one nonlinear tree-based model will achieve lower mean absolute scaled error than both the seasonal-naive and elastic-net models. Model selection will be based on repeated out-of-sample forecasts rather than training fit.

Research Question 4 (RQ4)

RQ4: Which variables contribute most to the preferred model, and does forecast performance differ materially across the five CMAs? The hypothesis is that lagged permit values and seasonal variables will have the greatest global importance and that forecast-error distributions will differ across CMAs because regional construction markets have different scales and volatility. The analysis will identify regions where forecast uncertainty is too high for direct operational use.

Feasibility, Measurability, and Ethical Fit

The proposed scope is achievable within 10 weeks because the outcome and primary predictors are already available as documented monthly government series, no primary data collection is required, and the model set is capped at four families. Each research question maps to observable variables, a defined hypothesis, and a planned statistical or machine-learning solution. The main technical risks are inconsistent category labels, revisions to historical values, outliers from unusually large projects, and dependence among monthly observations. These risks will be managed through a frozen data snapshot, a written data dictionary, time-aware validation, robust error metrics, sensitivity analyses, and clear limitations. Ethical risk is low because the analysis uses aggregate public data and does not predict individual conduct or eligibility.

Ten-Week Completion Plan

Table 1

Planned Capstone Schedule and Deliverables

Week	Primary activities	Evidence of completion
1	Finalize scope; download and freeze official data; verify categories and dates.	Source manifest, raw-data hashes, final RQs.
2	Complete focused literature review; create data dictionary and repository structure.	Literature matrix, README, dictionary draft.
3	Clean and join data; engineer lags and calendar features; document missingness.	Reproducible processed dataset and audit notebook.
4	Conduct exploratory analysis and answer RQ1.	Trend, seasonality, ACF, volatility, break diagnostics.
5	Fit seasonal-naïve and dynamic/elastic-net baselines; answer RQ2.	Baseline report and incremental-value analysis.
6	Fit random forest and XGBoost with time-aware tuning.	Tuned model objects and validation log.
7	Complete rolling-origin comparison and answer RQ3.	Metric table, confidence intervals, DM tests.
8	Perform SHAP, regional reliability, and sensitivity analysis for RQ4.	Driver plots, regional error profiles, robustness checks.
9	Develop recommendations; draft complete paper and visual summary.	Full draft, management interpretation, limitations.
10	Reproduce results from clean environment; APA review; final revisions.	Final PDF, public repository, presentation materials.

Note. The schedule reserves the final two weeks for interpretation, writing, reproducibility, and quality assurance rather than treating writing as an afterthought.

Sample Size Calculation

Sampling Unit and Power-Analysis Rationale

The Building Permits Survey is conducted as a census of the target population of Canadian permit-issuing municipalities rather than as a sample survey (Statistics Canada, 2026b). Consequently, the capstone does not recruit participants or draw a new probability sample. For this study, sample size refers to the number of usable CMA-month analytic observations and the number of paired out-of-sample forecast errors. Conventional power

calculations are therefore planning tools for detecting relationships and model-performance differences, not corrections for government sampling error. The calculations below use a two-sided alpha of .05 and statistical power of .80. Medium or small-to-medium effects were selected to avoid designing the project only to detect large, optimistic effects.

Table 2

Minimum Sample Size by Research Question

Research question	Method used	Key parameters	Minimum N
RQ1	Correlation / Fisher z	alpha = .05; power = .80; expected r = .30; two-sided	85 CMA-months
RQ2	Multiple regression F test	alpha = .05; power = .80; f-squared = .15; 10 tested predictors	118 CMA-months
RQ3	Paired forecast-error t test	alpha = .05; power = .80; standardized paired effect d = .35	67 paired forecasts
RQ4	One-way ANOVA	alpha = .05; power = .80; Cohen f = .25; 5 CMA groups	196 total (about 40 per CMA)

Note. Power values are planning assumptions. Final inference will account for temporal dependence through rolling-origin evaluation, block bootstrap confidence intervals, and heteroskedasticity-and-autocorrelation-consistent standard errors where applicable.

RQ1 Calculation

RQ1 requires enough observations to detect a medium association between a lagged or seasonal predictor and next-month permit value. Using Fisher's z transformation for a two-sided correlation test, the required sample is calculated as follows:

$$N = 3 + [(Z_{(1-\alpha/2)} + Z_{(1-\beta)}) / \text{atanh}(r)]^2$$

$$N = 3 + [(1.96 + 0.842) / \text{atanh}(0.30)]^2 = 84.93, \text{ rounded up to } 85$$

The planned analytic dataset will exceed this requirement within each major analysis and substantially exceed it when the five regional series are pooled.

RQ2 Calculation

RQ2 will be evaluated using a dynamic regression specification with permit lags, seasonal terms, region indicators, and macroeconomic predictors. A medium overall incremental effect of $f\text{-squared} = .15$, 10 tested predictors, $\alpha = .05$, and power = .80 requires approximately 118 observations under a fixed-model multiple-regression F test. This threshold is conservative for the planned model because not every coefficient must be interpreted separately; the central test is whether the macroeconomic block improves fit and out-of-sample accuracy.

RQ3 Calculation

RQ3 compares absolute or scaled errors from two models for the same forecast origin. A paired design is more efficient than an independent-groups design because each model is evaluated on the same observed month and region. For a small-to-medium standardized paired difference of $d = .35$, $\alpha = .05$, and power = .80, the minimum is 67 paired forecasts. Rolling-origin evaluation beginning after an initial 36-month training window is expected to produce more than 300 CMA-month forecasts, allowing both pooled and regional comparisons. Because neighboring forecast errors may be autocorrelated, significance will be confirmed with the Diebold-Mariano test and a moving-block bootstrap rather than relying only on a conventional paired t test (Diebold & Mariano, 1995).

RQ4 Calculation

RQ4 tests whether error differs across five CMAs. A one-way analysis of variance with five groups, Cohen's $f = .25$, $\alpha = .05$, and power = .80 requires approximately 196 total observations, or about 40 observations per CMA. The planned design provides 101 raw months per CMA and an estimated 89 analysis-ready observations per CMA after creating a 12-month lag and next-month target. Thus, even after limited missing-data exclusions, the regional comparison should retain adequate power.

Final Planned Sample

The maximum minimum requirement across the four calculations is 196 observations. The final planned sample is approximately 445 analysis-ready CMA-months: 89 observations per CMA across five CMAs after accounting for the 12-month lag and one-month-ahead target. This exceeds the maximum calculated minimum by more than twofold. The exact final N will be reported after the data snapshot is downloaded and all filters are applied. No synthetic records will be created to reach the target sample.

Data Description

Public Data Sources and Access

All study data will be obtained from government-operated, open-access sources. The source files will be downloaded once, assigned checksums, and stored in the repository through either a small raw file or a scripted download manifest, depending on the source terms and file size. The repository README will record the extraction date, table filters, transformations, software versions, and instructions required to rebuild the processed dataset.

Table 3

Government Data Sources

Source	Planned variables	Coverage / frequency	Public URL
Statistics Canada Table 34-10-0292-01	Residential permit value, geography, adjustment/value type, structure, type of work	January 2018-May 2026; monthly	https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3410029201
Statistics Canada Building Permits Survey metadata	Definitions, target population, collection design, accuracy and methodology	Active monthly program	https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&Id=1584733
Bank of Canada Valet	Target for the overnight rate	Daily source aggregated to monthly	https://www.bankofcanada.ca/valet/observations/V39079/csv?start_date=2018-01-01

series V39079		mean / end- of-month	
Statistics Canada Table 14- 10-0287- 01	Ontario unemployment rate, seasonally adjusted	Monthly	https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1410028701

Note. A direct bulk CSV for Table 34-10-0292-01 is also available at <https://www150.statcan.gc.ca/n1/en/tbl/csv/34100292-eng.zip>. URLs will be rechecked at final submission because government table identifiers and release schedules can change.

Expected Data Volume and Selection Rules

The primary cube will be filtered to the five selected CMAs, monthly frequency, total residential structures, total type of work, and seasonally adjusted current-dollar values. The target period contains 101 months. After sorting by CMA and date, the next-month target will be created with a one-period lead, and lags through 12 months will be computed only from prior information. The first 12 months and final row of each regional series will therefore be unavailable for the complete model matrix, leaving an estimated 88-89 observations per CMA depending on the exact publication cutoff. The joined policy-rate and unemployment series are expected to be complete; any missing source value will be documented rather than automatically interpolated.

Permit values will be inspected for zeros, negative revisions, suppression symbols, unit scaling, and exceptional projects. The primary modeling target will use $\log(1 + \text{value})$ to reduce extreme right skew, while all reported forecasts will be inverse-transformed to Canadian-dollar values. Because retransformation can be biased, the study will compare direct inverse transformation with a smearing correction. Raw current-dollar results will be supplemented by a constant-dollar sensitivity analysis where the Statistics Canada value-type dimension supports it.

Data Dictionary

Table 4 summarizes the core analytic fields. Appendix A provides the complete planned data dictionary, including provenance, type, unit, role, allowable values, transformation, and leakage status. The final dictionary will be generated from the analysis code so that the report and repository cannot silently diverge.

Table 4

Core Analytic Data Dictionary

Variable	Type / unit	Role	Operational definition
permit_value_t	Continuous; \$000s	Outcome history	Seasonally adjusted residential permit value for CMA in month t.
permit_value_t_plus_1	Continuous; \$000s	Forecast target	Residential permit value in the next calendar month.
lag_1, lag_3, lag_6, lag_12	Continuous; \$000s	Predictors	Prior permit values aligned within each CMA.
rolling_mean_3 / rolling_sd_3	Continuous	Predictors	Three-month prior-window level and volatility; excludes current target.
month_sin / month_cos	Continuous [-1,1]	Predictors	Cyclical encoding of month of year.
overnight_rate	Percent	Predictor	Monthly policy-rate summary from series V39079.
ontario_unemployment_rate	Percent	Predictor	Seasonally adjusted provincial unemployment rate.
cma	Categorical	Grouping / predictor	Toronto, Hamilton, Kitchener-Cambridge-Waterloo, London, or Windsor.

Reproducibility and GitHub Access

The accompanying public project repository is available at <https://github.com/dgupta91114/ontario-building-permit-forecasting> for advisor and course-team access. The repository will contain a README, environment specification, download scripts, a data manifest, notebooks mapped to RQ1-RQ4, reusable source modules, output tables and figures, the data dictionary, and a reproducibility log. Secrets, access tokens, or

private credentials will never be committed. Figure 2 in Appendix B presents the planned folder structure.

Data Ethics and Governance

The source data are aggregated, non-personal government statistics. The project will not infer individual financial status, evaluate permit applicants, or rank municipalities as good or bad performers. Results will be described as forecasts of published permit values, not forecasts of completed housing units or proof of causal policy effects. The processed data will preserve source flags for estimates that are revised, suppressed, or terminated. Model cards will disclose the study period, intended users, unsupported uses, known limitations, and uncertainty. These steps are important because open government data can still be misinterpreted or used outside the context for which they were collected.

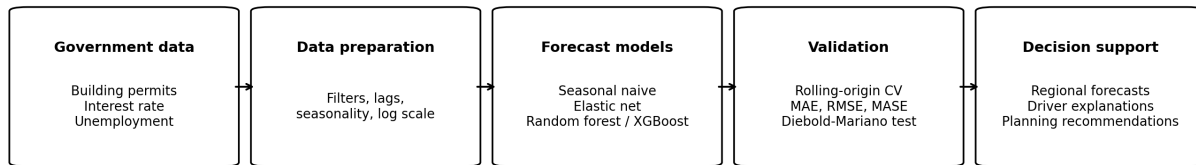
Analytic Approach

Overall Design

The study will use a retrospective panel time-series forecasting design. The analytic workflow is shown in Figure 1. The same data-preparation function will be used for every model, and all transformations will be fit using training data only. Random train-test splitting will not be used because it would allow future patterns to influence earlier forecasts. Instead, an expanding-window rolling-origin procedure will mimic repeated real-world monthly forecasting and preserve chronology (Bergmeir et al., 2018; Tashman, 2000).

Figure 1

Conceptual Framework for the Proposed Forecasting Study



Note. Government data are transformed into leakage-safe features, compared across benchmark and machine-learning models, validated through rolling forecast origins, and translated into region-specific planning guidance.

Preprocessing and Quality Checks

Data preparation will begin with schema validation, type conversion, duplicate-key checks, unit verification, and confirmation that each CMA-month has no more than one selected permit observation. The project will generate a complete month-by-CMA index and explicitly report missing combinations. Extreme values will not be removed merely because they are large; major permits are real economic events. Instead, the primary log-transformed model will be supplemented by robust-loss and winsorization sensitivity analyses. Augmented Dickey-Fuller and KPSS tests, autocorrelation plots, and seasonal decomposition will guide transformation choices without allowing an in-sample diagnostic to replace out-of-sample evaluation.

The initial training window will contain at least 36 months. At each forecast origin, models will be trained on all available prior months and evaluated on the next month. Hyperparameters will be selected within nested or prequential training folds and then locked before the final comparison. Region identifiers will be included as fixed features for pooled models. A sensitivity analysis will fit separate regional baselines to determine whether pooling improves accuracy for smaller CMAs.

Solution to RQ1: Temporal Structure

RQ1 will be addressed through descriptive statistics, indexed line plots, seasonal subseries plots, seasonal-trend decomposition using locally estimated scatterplot smoothing, autocorrelation and partial-autocorrelation functions, and dynamic regression. The statistical model will regress log next-month permit value on lagged permit values, harmonic seasonal terms, a linear time trend, CMA indicators, and prespecified break indicators. Newey-West heteroskedasticity-and-autocorrelation-consistent standard errors will be used for coefficient inference when residual dependence remains (Newey & West, 1987). The primary outputs will be the strength of annual seasonality, the persistence of lagged values, variance differences among CMAs, and a transparent assessment of whether the pandemic period materially altered the series.

RQ1 will be considered supported when the joint seasonal or lag block is statistically significant at $\alpha = .05$ and when its inclusion improves rolling-origin error relative to an intercept-and-region model. Statistical significance alone will not be interpreted as operational importance; standardized coefficients, partial R-squared, and changes in mean absolute error will also be reported.

Solution to RQ2: Incremental Value of Macroeconomic Predictors

RQ2 will compare two otherwise identical elastic-net dynamic regression models. Model A will use only permit history, calendar variables, trend, and CMA indicators. Model B will add the target overnight rate and Ontario unemployment rate, each considered contemporaneously and with one- to three-month lags that would have been available at the forecast origin. The added block will be evaluated with a nested-model test where assumptions are defensible, coefficient stability, and most importantly the percentage change in rolling-origin MAE and MASE. Because the project is predictive, it will follow Shmueli's (2010) distinction between explanatory association and forecasting performance.

The solution threshold is a prespecified reduction of at least 5% in pooled MAE without a material deterioration in three or more CMAs. If macroeconomic variables do not

meet this threshold, the correct conclusion will be that the added complexity is not justified for this horizon and period. This outcome would still answer the research question and strengthen the decision framework by preventing unnecessary model inputs.

Solution to RQ3: Model Comparison

RQ3 will compare four model families. The seasonal-naive model will forecast the value from the same month one year earlier and serves as the minimum credible benchmark. Elastic net will provide an interpretable regularized linear model. Random forest will capture nonlinear interactions and is robust to many distributional assumptions (Breiman, 2001). Extreme gradient boosting will sequentially fit trees to residual error and provides strong performance on structured data (Chen & Guestrin, 2016; Friedman, 2001). The project will not add deep neural networks because the available time series is too short to justify their complexity within the capstone timeline.

Each model will generate one-month-ahead forecasts over identical rolling origins. Primary ranking will use pooled MASE, followed by MAE and root mean squared error. Regional metrics, error dispersion, and prediction-interval coverage will be reported to prevent a large region from masking poor performance elsewhere. The Diebold-Mariano test will compare loss differentials between the leading model and each benchmark (Diebold & Mariano, 1995). A moving-block bootstrap will produce confidence intervals that respect short-range dependence. The selected model must improve on the seasonal-naive benchmark and demonstrate stable error rather than winning because of a few months.

Solution to RQ4: Explainability and Regional Reliability

RQ4 will use permutation importance and SHapley Additive exPlanations (SHAP) for the preferred tree-based model, or standardized coefficients if elastic net is selected. SHAP values allocate a prediction among features using a consistent game-theoretic framework and can support global and local explanations (Lundberg & Lee, 2017). The study will report the top drivers, the direction and nonlinearity of selected effects, and representative high- and low-

forecast months. Explanations will be computed only on held-out or rolling-origin predictions to reduce the temptation to explain overfit training patterns.

Regional reliability will be assessed with CMA-specific MASE, median absolute error, error bias, interval coverage, and one-way or robust analysis of variance on absolute scaled errors. Post hoc comparisons will use multiplicity correction. A model may be recommended for pooled strategic monitoring even if one CMA requires caution. The final decision rule will label a regional forecast as operationally acceptable only when MASE is below 1.0, median bias is limited, and interval coverage is reasonably close to the nominal level.

Evaluation Metrics and Formulae

Forecast evaluation will use multiple complementary metrics because no single measure captures all operational concerns. Let y_i denote the observed permit value, $\hat{y}_i$ the forecast, $e_i = y_i - \hat{y}_i$ the error, and n the number of forecasts.

$$MAE = (1/n) * \sum |y_i - \hat{y}_i|$$

Mean absolute error (MAE) expresses the typical error in the original monetary scale and is the most interpretable metric for planners.

$$RMSE = \sqrt{(1/n) * \sum (y_i - \hat{y}_i)^2}$$

Root mean squared error (RMSE) penalizes large misses more heavily and therefore highlights models that fail during unusually large permit months.

$$sMAPE = (100/n) * \sum [2|y_i - \hat{y}_i| / (|y_i| + |\hat{y}_i|)]$$

Symmetric mean absolute percentage error (sMAPE) is scale free but will be interpreted cautiously when observed and forecast values are close to zero.

$$MASE = MAE_{\text{model}} / \text{mean}(|y_t - y_{(t-12)}|)$$

Mean absolute scaled error (MASE) compares the candidate model with a seasonal-naive benchmark. A value below 1.0 indicates improvement over that benchmark (Hyndman & Koehler, 2006).

$$\text{Bias} = (1/n) * \sum (\hat{y}_i - y_i)$$

Bias identifies systematic over- or underforecasting. For interval forecasts, prediction interval coverage probability will be calculated as the proportion of observed values that fall within the nominal interval. All metrics will be reported on the inverse-transformed original-dollar scale, with log-scale diagnostics retained for model development.

Robustness and Sensitivity Analyses

The study will repeat the principal comparison under several prespecified variations: current versus constant dollars; with and without the pandemic-period observations; pooled versus separate CMA models; standard versus robust treatment of extreme months; and alternate initial training-window lengths. Residual diagnostics will include autocorrelation, heteroskedasticity, and error-versus-level plots. Model conclusions will be described as robust only if the preferred family remains competitive across most specifications. This guards against a high grade being driven by a single favorable split or a single metric.

Recommendation and Application

Target Users

The primary end users are municipal and provincial economic-development analysts, residential builders and developers, building-material suppliers, construction workforce planners, lenders, and industry associations. Each user requires a different level of detail, but all benefit from a transparent estimate of near-term construction intentions and a clear statement of uncertainty. The project is designed as a decision aid rather than an automated decision-maker.

Table 5

Proposed Users, Decisions, and Benefits

End user	Decision supported	Expected benefit
Municipal / provincial analysts	Monitor regional momentum; prioritize further investigation.	Earlier, reproducible signal of changes in planned residential activity.

Builders and developers	Scenario planning, pipeline timing, and competitive context.	Benchmark internal expectations against public regional evidence.
Material suppliers	Inventory and distribution planning.	Reduce stock-outs or excess inventory caused by purely reactive planning.
Workforce planners and colleges	Training, hiring, and apprenticeship capacity discussions.	Identify regions where planned activity may create near-term labour pressure.
Lenders and risk teams	Portfolio monitoring and market context.	Add a transparent external indicator without inferring individual borrower risk.

Planned Recommendation Logic

The final recommendation will depend on the evidence rather than presuming that machine learning must win. If a tree-based model produces a meaningful and stable reduction in MASE, the study will recommend an explainable monthly monitoring tool with region-specific reliability flags. If elastic net performs similarly, the simpler model will be recommended because it is easier to audit, reproduce, and communicate. If no candidate consistently beats seasonal naive, the recommendation will be to retain the baseline and use the analysis to identify where additional predictors or longer histories are needed. This outcome-sensitive logic aligns model choice with governance and operational value rather than novelty.

The deliverable will include a concise regional scorecard showing the next-month forecast, an uncertainty interval, recent trend, benchmark comparison, major drivers, and a reliability label. End users will be instructed not to interpret permit value as completed construction, housing supply, or future price. Large projects can cause legitimate spikes, permits can be revised, and economic relationships may change. A monthly refresh and at least annual model review will therefore be recommended.

Anticipated Contribution and Limitations

The doctoral-level contribution is a reproducible decision framework that integrates official construction and macroeconomic data, statistically powered research questions, time-aware forecast validation, explainable machine learning, regional reliability analysis, and

explicit deployment rules. The study extends beyond selecting the algorithm with the smallest error by asking when added data are useful, whether gains persist across regions, and how uncertainty should constrain application.

Limitations include the relatively short current-table history beginning in 2018, revisions to government data, dependence among observations, aggregation at the CMA level, potential differences in municipal permitting practices, and the fact that permit value measures construction intentions rather than completed work. The interest-rate and unemployment variables are common to all CMAs within a month, which limits identification of regional macroeconomic differences. The results will apply to the selected period, geography, and one-month horizon and should not be generalized automatically to other provinces or longer horizons. These limitations will be treated as boundaries on interpretation, not hidden after model fitting.

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Appendix A

Planned Full Data Dictionary

This appendix is referenced in the Data Description section. Field names may be adjusted after the source cube is downloaded, but every final field will retain the documented source, unit, role, and transformation.

Table 6

Planned Full Data Dictionary

Field	Source	Type / unit	Role	Definition / treatment
ref_date	Statistics Canada	Date; month	Key	Reference month converted to first day of month.
cma	Statistics Canada	Categorical	Key / predictor	One of five selected Ontario CMAs.
dguide	Statistics Canada	String	Audit key	Official geographic identifier retained for reproducibility.
seasonal_adjustment	Statistics Canada	Categorical	Filter	Seasonally adjusted series selected for primary analysis.
value_type	Statistics Canada	Categorical	Filter / sensitivity	Current dollars primary; constant dollars sensitivity.
type_of_structure	Statistics Canada	Categorical	Filter	Residential structures, total.
type_of_work	Statistics Canada	Categorical	Filter	Types of work, total.
permit_value_t	Statistics Canada	Continuous; \$000s	Predictor history	Published permit value for month t.
permit_value_t_plus_1	Derived	Continuous; \$000s	Target	Within-CMA one-month lead of permit value.
log_target	Derived	Continuous	Target transform	$\log(1 + \text{permit_value_t_plus_1})$.
lag_1	Derived	Continuous; \$000s	Predictor	Permit value at t-1.
lag_3	Derived	Continuous; \$000s	Predictor	Permit value at t-3.

lag_6	Derived	Continuous; \$000s	Predictor	Permit value at t-6.
lag_12	Derived	Continuous; \$000s	Predictor / benchmark	Permit value at t-12.
rolling_mean_3	Derived	Continuous; \$000s	Predictor	Mean of prior three observed months, excluding target.
rolling_sd_3	Derived	Continuous; \$000s	Predictor	Standard deviation of prior three observed months.
month_sin	Derived	Continuous	Predictor	$\sin(2\pi \cdot \text{month}/12)$.
month_cos	Derived	Continuous	Predictor	$\cos(2\pi \cdot \text{month}/12)$.
time_index	Derived	Integer months	Predictor	Months elapsed since study start.
pandemic_indicator	Derived	Binary	Sensitivity predictor	Prespecified disruption window; final dates documented.
overnight_rate	Bank of Canada	Percent	Predictor	Monthly summary of V39079 available at forecast origin.
overnight_rate_lag_1	Derived	Percent	Predictor	Prior-month target overnight rate.
ontario_unemployment_rate	Statistics Canada	Percent	Predictor	Seasonally adjusted provincial unemployment rate.
unemployment_lag_1	Derived	Percent	Predictor	Prior-month unemployment rate.
source_status	Statistics Canada	Categorical	Quality flag	Source symbol/status retained and audited.
train_fold	Derived	Integer	Validation control	Rolling-origin fold identifier; never used as predictor.

Appendix B

Planned GitHub Repository and Reproducibility Structure

This appendix is referenced in the Reproducibility and GitHub Access section. The accompanying public repository is available at <https://github.com/dgupta91114/ontario-building-permit-forecasting> and contains the organized project structure shown below.

Figure 2

Planned Public Repository Structure

Planned public GitHub repository

<https://github.com/REPLACE-WITH-USERNAME/ontario-building-permit-forecasting>

```

ontario-building-permit-forecasting/
|-- README.md
|-- LICENSE
|-- requirements.txt
|-- data/
|   |-- raw/           # Source files or download manifest
|   |-- interim/       # Cleaned intermediate files
|   |-- processed/     # Analysis-ready panel dataset
|-- notebooks/
|   |-- 01_data_audit.ipynb
|   |-- 02_eda_rq1.ipynb
|   |-- 03_dynamic_regression_rq2.ipynb
|   |-- 04_model_comparison_rq3.ipynb
|   |-- 05_explainability_rq4.ipynb
|-- src/
|   |-- download_data.py
|   |-- prepare_data.py
|   |-- features.py
|   |-- train.py
|   |-- evaluate.py
|-- reports/
|   |-- figures/
|   |-- tables/
|-- docs/
|   |-- data_dictionary.csv
|   |-- reproducibility_log.md

```

Note. The raw-data directory will contain either source files permitted for redistribution or a manifest and scripted downloader. The README will include exact reproduction instructions and the final public URL.

Appendix C

Sample Size Assumptions and Decision Rules

The following assumptions will be frozen before outcome analysis: $\alpha = .05$, power = .80, two-sided tests, correlation $r = .30$ for RQ1, regression f-squared = .15 for RQ2, paired-error effect $d = .35$ for RQ3, and ANOVA $f = .25$ for RQ4. The maximum minimum N is 196. The estimated analytic N is approximately 445 CMA-months, subject to final source availability and lag construction.

Primary model-selection rule: select the simplest model that (a) achieves pooled MASE below 1.0, (b) has lower pooled MAE than seasonal naive with uncertainty supporting a genuine improvement, (c) does not materially degrade performance in three or more CMAs, and (d) passes reproducibility checks. If no candidate meets these conditions, retain seasonal naive and report the negative result.

Primary RQ2 decision rule: conclude that macroeconomic predictors add practical value only when they reduce pooled rolling-origin MAE by at least 5% and do not introduce substantial regional instability. This threshold prevents a statistically detectable but operationally negligible difference from being overstated.